

Step-by-Step Instructions

Step 1. Open the Website

1. Go to <https://alfalfa.gdsl.org/> in your browser.
2. You'll see the homepage with the **Login** and **Sign Up** buttons at the top-right corner.

AlfAdvisor[Documentation](#)[Login](#)[Sign Up](#)

Welcome to web-based cyber-platform to estimate alfalfa yield and quality to support harvest scheduling

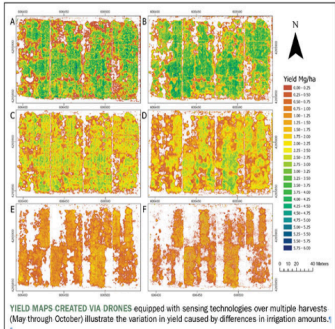
This is a free public cyberplatform (AlfAdvisor) which will provide alfalfa growers with economically optimum harvesting recommendations. The platform will include explicit consideration of alfalfa growth and quality changes, drying rates and weather-related risks incurred during harvest.

Objectives

Objective 1:
Develop models to estimate alfalfa yield, quality and drying rate in real-time by combining satellite imagery and environmental factors to inform harvesting time.

Objective 2:
Develop a decision-making model to provide economically optimum harvest scheduling during the growing season.

Objective 3:
Create a web-based cyber-platform to disseminate tools for free public use



YIELD MAPS CREATED VIA DRONES equipped with sensing technologies over multiple harvests (May through October) illustrate the variation in yield caused by differences in irrigation amounts.¹

Step 2. Log in or Sign Up

1. Click **Login** to access your account.
2. If you don't have an account, click **Sign Up** and fill in your information.

Sign Up

First Name:

Last Name:

Email:

Password:

Confirm Password:

Already registered?
[Login](#)

Login

Email:

Password:

Need an Account?
[Sign Up](#)

Step 3. Add Your Farm

1. Click **Add Farm**, then select **Place Marker**.
2. Place the marker roughly at the center of your research area.

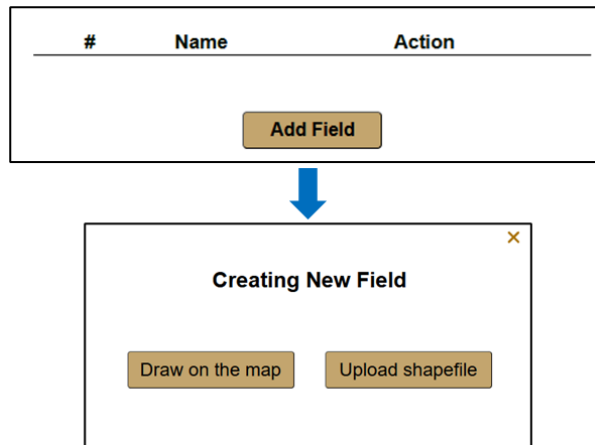
3. Enter a farm name and click **Add Farm**. You will see that the farm has been created successfully.
4. If you already have a farm, you can **open**, **edit**, or **delete** it.

The sequence of screenshots illustrates the process of adding a farm to a map application:

- Step 1:** A map of North America is shown. A red box highlights the **Add Farm** button in the top right corner, labeled with a red **1**.
- Step 2:** A map of North America is shown. A red box highlights a location in the top left corner, labeled with a red **2**.
- Step 3:** A map of North America is shown. A red box highlights a **Farm Name** input field and an **Add Farm** button, labeled with a red **3**.
- Step 4:** A map of North America is shown. A red box highlights a table and an **Add Farm** button, labeled with a red **4**. The table has columns **#**, **Name**, and **Action**. It contains one row with **# 1**, **Name Text**, and **Action** buttons **Open**, **Edit**, and **Delete**.

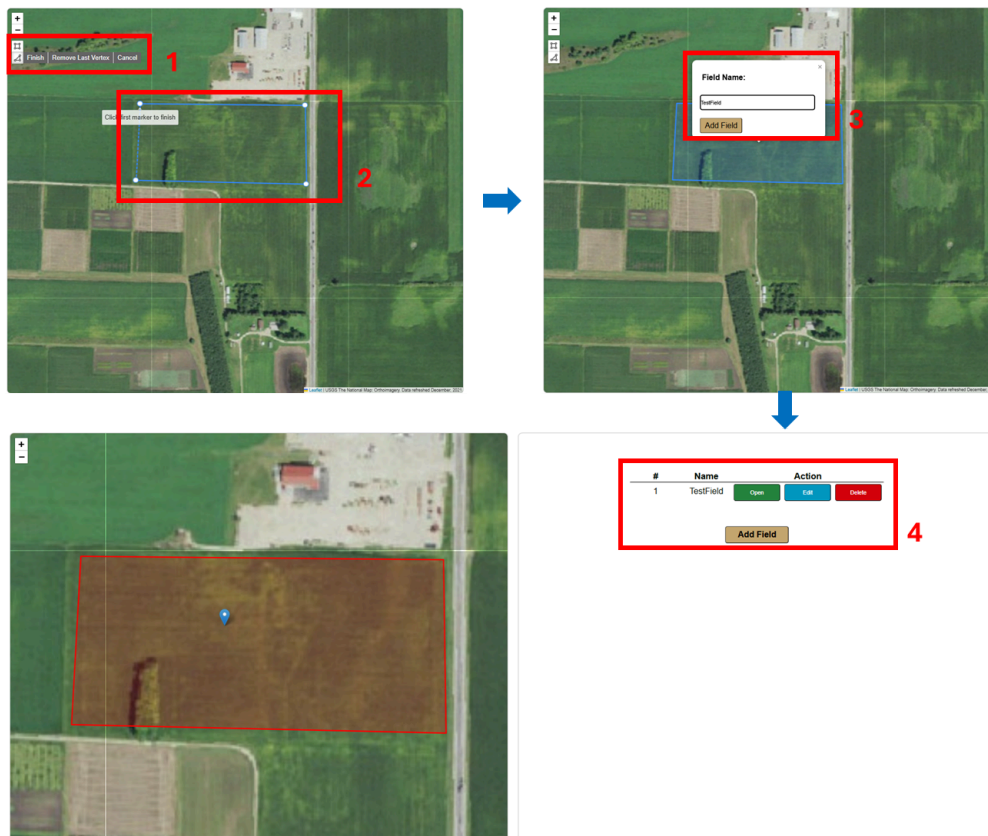
Step 4. Add Your Alfalfa Field

1. Open the farm where you want to add the field, then click **Add Field**. You will see two options: **Draw on the Map** and **Upload Shapefile**.



2. If you choose “**Draw on the Map**”:

Zoom into your field area, then use the **Draw Rectangle** or **Draw Polygon** tool on the left to manually outline your field boundary. After finishing, set a Field Name and save it.



3. If you choose “**Upload Shapefile**”:

Click **Choose Files**, select your shapefile from the pop-up window, then click **Upload**.

After the file is uploaded, enter the Field Name and click **Add Field**.

Upload all necessary files for field shapefile
(.shp, .shx, .dbf, .prj)

Choose Files

No file chosen

Upload

Enter Field Name:

Enter field name

Add Field



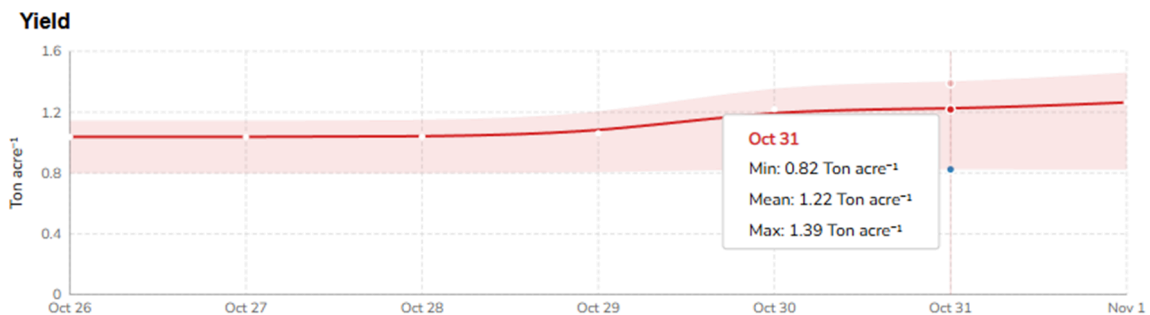
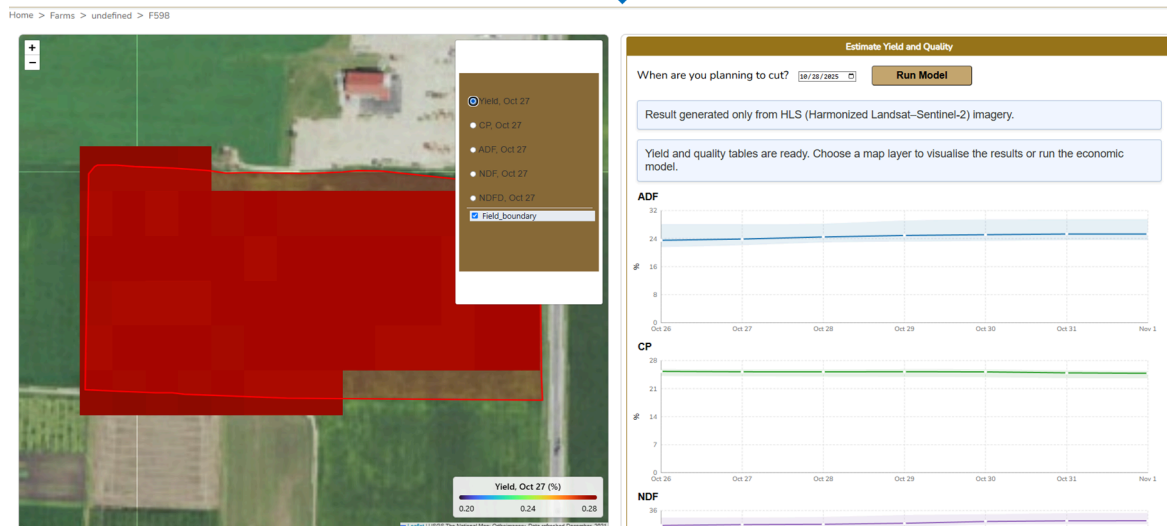
#	Name	Action
1	F598	Open Edit Delete

Add Field

4. If you already have a field, you can **open**, **edit**, or **delete** it.

Step 5. Estimate Yield and Quality

After opening a field, you will see the forecasted precipitation and temperature. Then, scroll to the Estimate Yield and Quality section. Set your **expected harvest date**, click **Run Model**, and wait a few moments. The system will generate the predicted yield and quality results for the next seven days.



Step 6. Run Economic Model

After generating yield and quality predictions, you can estimate the expected revenue.

- Go to the Run Economic Model section.
- Enter the required basic information.
- Click **Show Results** to view the estimated drying time and expected income for each of the next seven harvest days.

Run Economic Model

Market: Sell

Tedding: No

Initial Moisture Content(%): 90

Target Moisture Content(%): 10

Show Results

